

Abby Grace: Renewable Energy by U.S. States (2021-2023)

Example Solution 1

Overview

I looked at how the total energy use has changed accross all of the U.S. states from the years 2021 to 2023. I used the data that was provided in the data file and kept on the row which has the total renewable energy row for each year in all three of the different csv's. I also reshaped the data to a tibble containing all three years and all states for the total renewable energy use. The new tibble has three columns 'state', 'renew_use', and 'year'.

Part 0: libraries

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
library(readr)
library(ggplot2)
library(tibble)
library(stringr)
library(tidyr)
library(usmap)
```

Part 1: Defining Research Question

Chosen Question: How has the total renewable energy use changed from 2021-2023 across U.S. states?

Part 2: Data Preparation and Cleaning

```
#load files
use21 <- read_csv('/Users/abbygrace/Desktop/stat133/ev-power-abbygrace/data/total-use-2021.csv')
```

```
Rows: 5 Columns: 53
-- Column specification -----
Delimiter: ","
chr  (1): Energy_Source
dbl (52): AK, AL, AR, AZ, CA, CO, CT, DC, DE, FL, GA, HI, IA, ID, IL, IN, KS...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
use22 <- read_csv('/Users/abbygrace/Desktop/stat133/ev-power-abbygrace/data/total-use-2022.csv')
```

```
Rows: 5 Columns: 53
-- Column specification -----
Delimiter: ","
chr  (1): Energy_Source
dbl (52): AK, AL, AR, AZ, CA, CO, CT, DC, DE, FL, GA, HI, IA, ID, IL, IN, KS...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
use23 <- read_csv('/Users/abbygrace/Desktop/stat133/ev-power-abbygrace/data/total-use-2023.
```

```
Rows: 5 Columns: 53
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Energy_Source
```

```
dbl (52): AK, AL, AR, AZ, CA, CO, CT, DC, DE, FL, GA, HI, IA, ID, IL, IN, KS...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#need to make the column nanmes the same
```

```
use21 <- use21 |> rename_with(~ str_replace_all(tolower(.x), " ", "_"))
```

```
use22 <- use22 |> rename_with(~ str_replace_all(tolower(.x), " ", "_"))
```

```
use23 <- use23 |> rename_with(~ str_replace_all(tolower(.x), " ", "_"))
```

```
#make all energy source names consistently lowercase
```

```
use21$energy_source <- tolower(use21$energy_source)
```

```
use22$energy_source <- tolower(use22$energy_source)
```

```
use23$energy_source <- tolower(use23$energy_source)
```

```
#now use regular expressions to make all consistent
```

```
use21$energy_source <- str_replace_all(use21$energy_source, "total.*renew.*", "total_renewab.
```

```
use22$energy_source <- str_replace_all(use22$energy_source, "total.*renew.*", "total_renewab.
```

```
use23$energy_source <- str_replace_all(use23$energy_source, "total.*renew.*", "total_renewab.
```

```
#check they all look the same
```

```
head(use21)
```

```
# A tibble: 5 x 53
```

energy_source	ak	al	ar	az	ca	co	ct	dc	de
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 coal	18694	309791	216123	160299	2.82e4	252442	2880	0	4542
2 natural gas†	395590	739891	360545	484962	2.17e6	509970	305184	28336	82708
3 petroleum (btu)	261094	583042	328271	606862	2.96e6	497788	284788	18439	113641
4 nuclear	0	480115	141372	329868	1.72e5	0	179551	0	0
5 total_renewable~	9597	239817	89714	99266	8.10e5	103955	49306	2487	7150

```
# i 43 more variables: fl <dbl>, ga <dbl>, hi <dbl>, ia <dbl>, id <dbl>,
#   il <dbl>, `in` <dbl>, ks <dbl>, ky <dbl>, la <dbl>, ma <dbl>, md <dbl>,
#   me <dbl>, mi <dbl>, mn <dbl>, mo <dbl>, ms <dbl>, mt <dbl>, nc <dbl>,
#   nd <dbl>, ne <dbl>, nh <dbl>, nj <dbl>, nm <dbl>, nv <dbl>, ny <dbl>,
```

```
# oh <dbl>, ok <dbl>, or <dbl>, pa <dbl>, ri <dbl>, sc <dbl>, sd <dbl>,
# tn <dbl>, tx <dbl>, ut <dbl>, va <dbl>, vt <dbl>, wa <dbl>, wi <dbl>,
# wv <dbl>, wy <dbl>, us <dbl>
```

```
head(use22)
```

```
# A tibble: 5 x 53
  energy_source      ak      al      ar      az      ca      co      ct      dc      de
  <chr>          <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 coal consumption 18615 297654 211724 154007 3.00e4 233256      0      0    1846
2 natural-gas      437916 787300 398099 468038 2.13e6 524890 307212 30174 89674
3 petroleum (btu) 263335 578431 327813 594859 3.02e6 538413 302881 18000 112026
4 nuclear energy†      0 442093 149654 333738 1.84e5      0 172018      0      0
5 total_renewable~ 10410 232035 90825 101215 8.81e5 114917 49084 2622 7402
# i 43 more variables: fl <dbl>, ga <dbl>, hi <dbl>, ia <dbl>, id <dbl>,
# il <dbl>, `in` <dbl>, ks <dbl>, ky <dbl>, la <dbl>, ma <dbl>, md <dbl>,
# me <dbl>, mi <dbl>, mn <dbl>, mo <dbl>, ms <dbl>, mt <dbl>, nc <dbl>,
# nd <dbl>, ne <dbl>, nh <dbl>, nj <dbl>, nm <dbl>, nv <dbl>, ny <dbl>,
# oh <dbl>, ok <dbl>, or <dbl>, pa <dbl>, ri <dbl>, sc <dbl>, sd <dbl>,
# tn <dbl>, tx <dbl>, ut <dbl>, va <dbl>, vt <dbl>, wa <dbl>, wi <dbl>,
# wv <dbl>, wy <dbl>, us <dbl>
```

```
head(use23)
```

```
# A tibble: 5 x 53
  energy_source      ak      al      ar      az      ca      co      ct      dc      de
  <chr>          <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 coal_usage      18414 224926 180262 137885 2.87e4 204826      0      0    338
2 naturalgas      448087 775747 399566 537151 2.15e6 525446 304924 26236 84387
3 petroleum (btu) 270391 565754 327465 599712 3.00e6 514174 292864 17292 110721
4 nuclear-energy †      0 476392 156492 329474 1.85e5      0 142873      0      0
5 total_renewable~ 10087 222189 87277 108445 1.07e6 115061 48981 2795 8041
# i 43 more variables: fl <dbl>, ga <dbl>, hi <dbl>, ia <dbl>, id <dbl>,
# il <dbl>, `in` <dbl>, ks <dbl>, ky <dbl>, la <dbl>, ma <dbl>, md <dbl>,
# me <dbl>, mi <dbl>, mn <dbl>, mo <dbl>, ms <dbl>, mt <dbl>, nc <dbl>,
# nd <dbl>, ne <dbl>, nh <dbl>, nj <dbl>, nm <dbl>, nv <dbl>, ny <dbl>,
# oh <dbl>, ok <dbl>, or <dbl>, pa <dbl>, ri <dbl>, sc <dbl>, sd <dbl>,
# tn <dbl>, tx <dbl>, ut <dbl>, va <dbl>, vt <dbl>, wa <dbl>, wi <dbl>,
# wv <dbl>, wy <dbl>, us <dbl>
```

Part 3: Joining / Pivoting Datasets for Analysis

```
#now we want to add all 3 years in one table
#now we want to keep the row which is named "total_renewable_energy" and make our table longer
#we want the state, the total renewable energy use

renewable21 <- use21 |>
  filter(energy_source == "total_renewable_energy") |>
  pivot_longer(cols = - energy_source, names_to = "state", values_to = "renewable energy use")
  mutate(year = 2021)

renewable22 <- use22 |>
  filter(energy_source == "total_renewable_energy") |>
  pivot_longer(cols = - energy_source, names_to = "state", values_to = "renewable energy use")
  mutate(year = 2022)

renewable23 <- use23 |>
  filter(energy_source == "total_renewable_energy") |>
  pivot_longer(cols = - energy_source, names_to = "state", values_to = "renewable energy use")
  mutate(year = 2023)

renewable_energy_all <- bind_rows(renewable21, renewable22, renewable23)

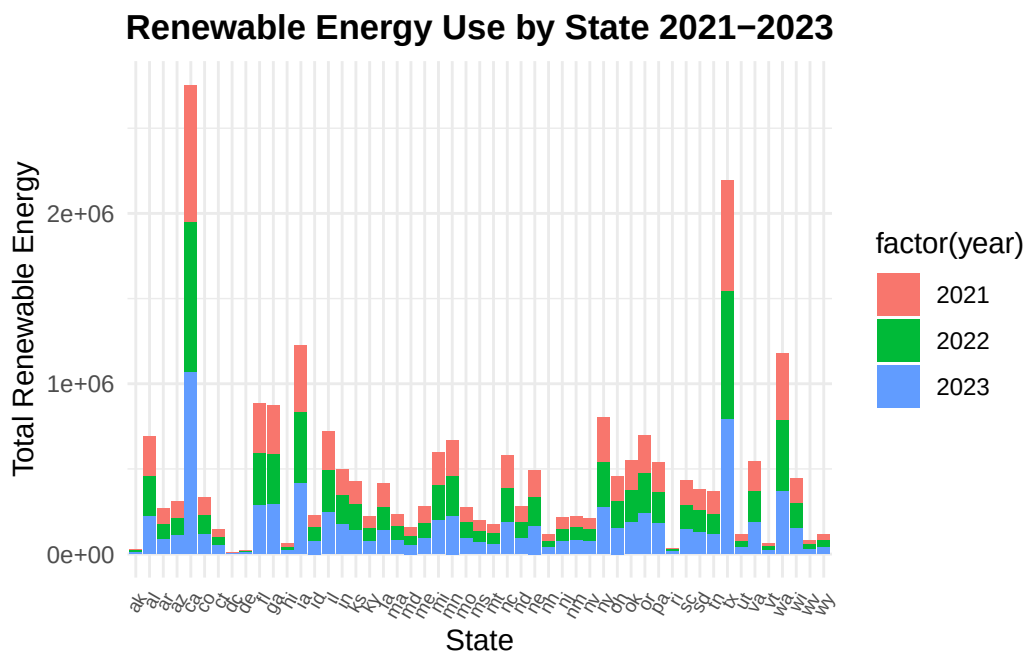
renewable_energy_all <- renewable_energy_all |>
  filter(state != "us") |>
  select(-energy_source) |>
  rename(renew_use = 'renewable energy use')

head(renewable_energy_all)
```

```
# A tibble: 6 x 3
  state renew_use year
  <chr>    <dbl> <dbl>
1 ak      9597  2021
2 al    239817  2021
3 ar     89714  2021
4 az     99266  2021
5 ca    810020  2021
6 co    103955  2021
```

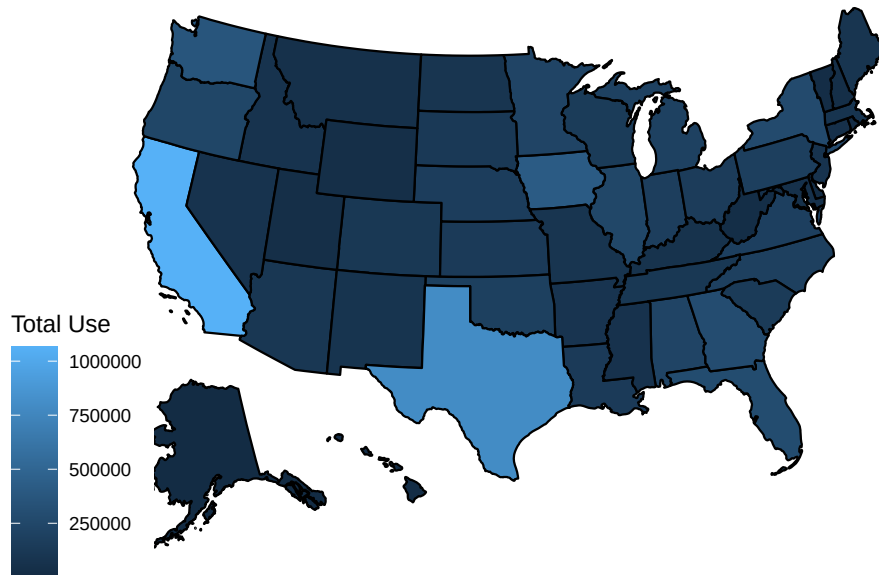
Part 4: Mapping Visualization

```
ggplot(renewable_energy_all, aes(x = state, y = renew_use, fill = factor(year))) +  
  geom_col() +  
  theme_minimal() +  
  labs(title = "Renewable Energy Use by State 2021-2023", x = 'State', y = 'Total Renewable  
  theme(  
    axis.text.x = element_text(angle = 60, hjust = 1, size = 7),  
    plot.title = element_text(hjust = 0.5, face = "bold"))
```



```
plot_usmap(data = filter(renewable_energy_all, year == 2023) |>  
  mutate(state = toupper(state)), values = 'renew_use', regions = "states") +  
  labs(title = "Renewable Energy Use by State (2023)", fill = "Total Use")
```

Renewable Energy Use by State (2023)



Part 5: Final Deliverable

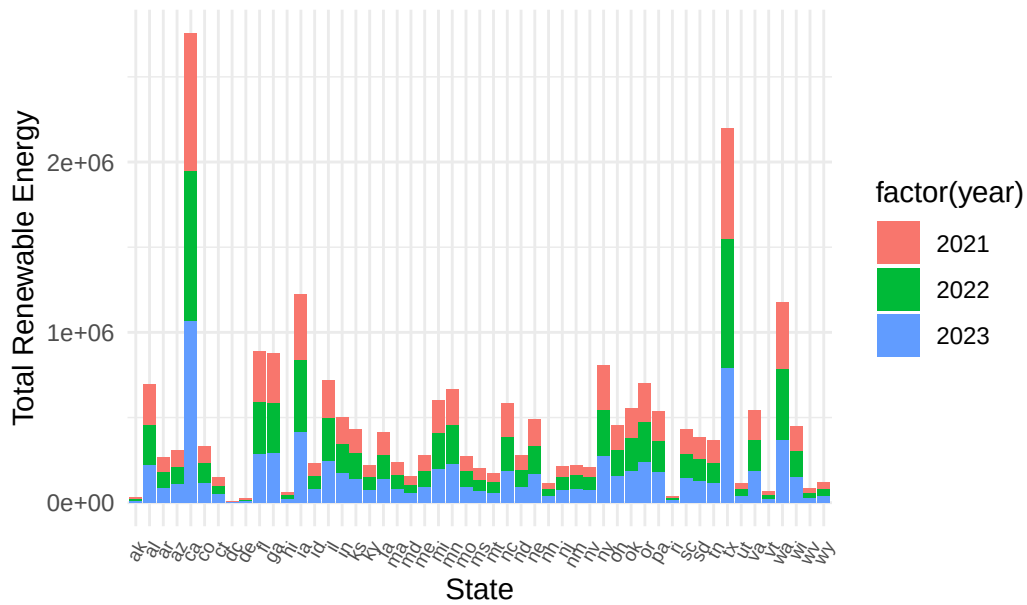
The map shows the total renewable energy use by state in 2023. The lighter shade of blue indicates more usage while the darker shade shows less use. I also added a bar chart which shows the total renewable energy across all U.S. states per year.

I noticed that states with the largest total renewable energy in 2023 were California and Texas. Many states show increase in renewable energy from 2021 to 2023 by the bar chart, which is a good sign! I think the map helps us visualize the comparison per state easier than the bar chart. The bar chart you have to look at the numbers as well as the states and compare the different bar sizes per year, which can be difficult to do.

Here are my plots

```
ggplot(renewable_energy_all, aes(x = state, y = renew_use, fill = factor(year))) +  
  geom_col() +  
  theme_minimal() +  
  labs(title = "Renewable Energy Use by State 2021-2023", x = 'State', y = 'Total Renewable') +  
  theme(  
    axis.text.x = element_text(angle = 60, hjust = 1, size = 7),  
    plot.title = element_text(hjust = 0.5, face = "bold"))
```

Renewable Energy Use by State 2021–2023



```
plot_usmap(data = filter(renewable_energy_all, year == 2023) |>
  mutate(state = toupper(state)), values = 'renew_use', regions = "states") +
  labs(title = "Renewable Energy Use by State (2023)", fill = "Total Use")
```

Renewable Energy Use by State (2023)

